

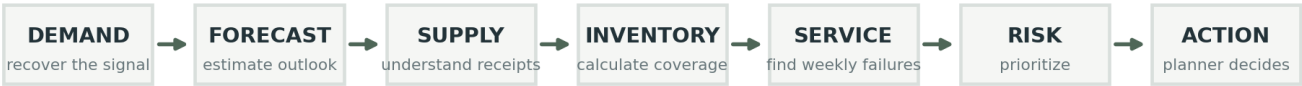
DemandIQ

How I would move from stockout-hidden sales to a governed 13-week planning decision

Mature-product planning simulation | Week x SKU x Channel | 9 forecast series

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My goal was not to build the most complex forecast. It was to show how I would structure an ambiguous planning problem, test the assumptions, quantify the trade-offs, and move the analysis toward a decision.



I treat forecasting as one step inside a larger planning decision.

13W BASE DEMAND	PORTFOLIO FILL	ACUTE WEEKLY RISK	FORWARD COVERAGE RISK
36.4K	98.8%	3 of 9	9 of 9
units	vs 92% target	series	below 2.5-week policy

Scope note: this is a synthetic portfolio simulation inspired by premium outdoor apparel planning. It is not Arc'teryx data and does not represent any company's internal process.

1. THE CHALLENGE

I started with the demand signal, not the model.

The planning question: how should I forecast demand when historical sales may themselves have been constrained by inventory?

Observed sales tell me what the business sold. During a stockout, they may not tell me what customers wanted to buy.

That creates a downstream risk:

- Inventory constrains availability and observed sales fall.
- Historical demand appears weaker than the underlying customer need.
- A forecast trained directly on those sales can learn the stockout as low demand.
- The resulting buy or receipt plan can become too conservative and recreate the service problem.

So I treated demand reconstruction as a planning problem before treating forecasting as a modeling problem.



I treat forecasting as one step inside a larger planning decision.

MY PLANNING PRINCIPLES

1. Separate demand from availability

Sales are not automatically demand when inventory constrained the customer's ability to buy.

2. Forecasts need operating context

A forecast is useful only when receipts, inventory coverage, service and execution feasibility are considered.

3. Look below aggregate KPIs

A healthy portfolio metric can hide localized weekly failures.

4. Keep judgment outside the model where appropriate

If supplier, transfer or expedite feasibility is not modeled, the system should escalate - not authorize.

Three decisions shaped the case study.

I designed the workflow around information a planner could realistically use, then kept synthetic truth and generator-only variables outside the forecast.

01

What should I forecast?

I used Seasonal Profile Imputation to reconstruct demand during censored periods and forecasted reconstructed_demand_units - not raw stockout-depressed sales.

95.68% of censored demand recovered in the simulation.

02

At what level should I plan?

I selected week x SKU x Channel, aggregating region first. This produced nine mature series: granular enough to expose channel risk without unnecessary fragmentation.

9 forecast series | 13-week horizon.

03

How should I choose the model?

I used 12 expanding-window backtest folds, WAPE as the primary metric and bias as a separate governance check. I favored the simplest defensible champion rather than complexity for its own sake.

Champion WAPE: 8.4%-21.7% | Bias: -3.1% to +2.4%.

Governance rule: true_demand_units exists only as hidden synthetic evaluation truth. It was never a forecast input. Generator-only drivers were also excluded from the modeling feature set.

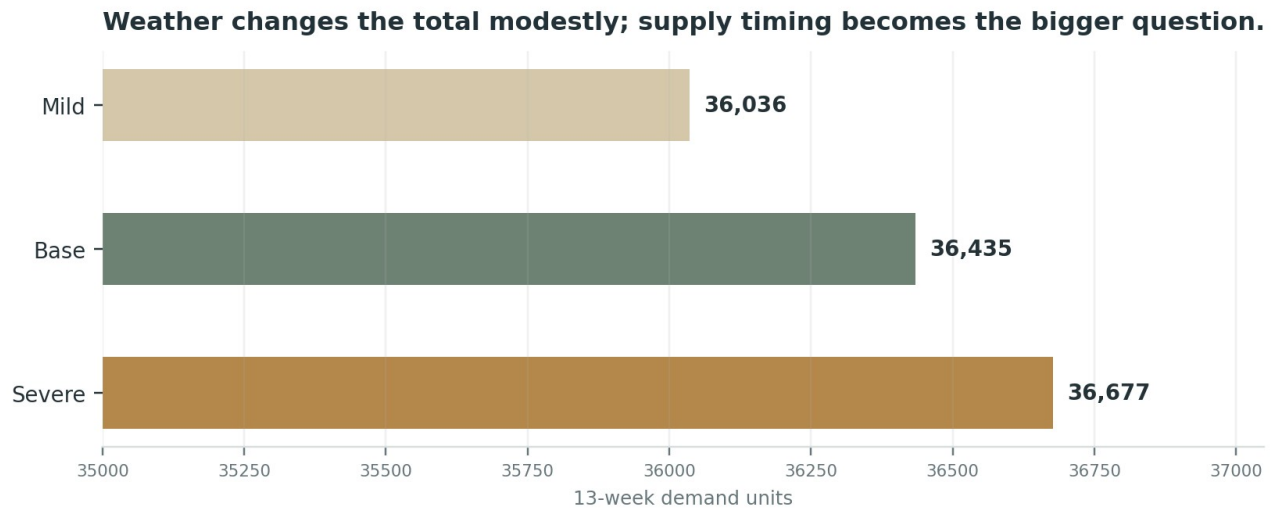
A model-selection example I would defend in an interview:

On IMH-001 / ECOM, SARIMA improved WAPE by only about 0.06 percentage points versus ETS. I retained ETS because I did not consider that marginal gain sufficient to justify additional complexity and maintainability cost.

Weather mattered less than I expected.

Supply timing mattered more.

I included weather as a demand-driver layer, but the analysis did not support making it the headline risk.



MILD	BASE	SEVERE	SCENARIO WIDTH
36,036	36,435	36,677	641
units	units	units	~1.8% of Base

My interpretation

The severe-weather case was only +0.67% above Base over the 13-week window. Rather than overselling the weather layer, I changed the emphasis of the analysis: the stronger near-term planning question was when supply landed relative to demand.

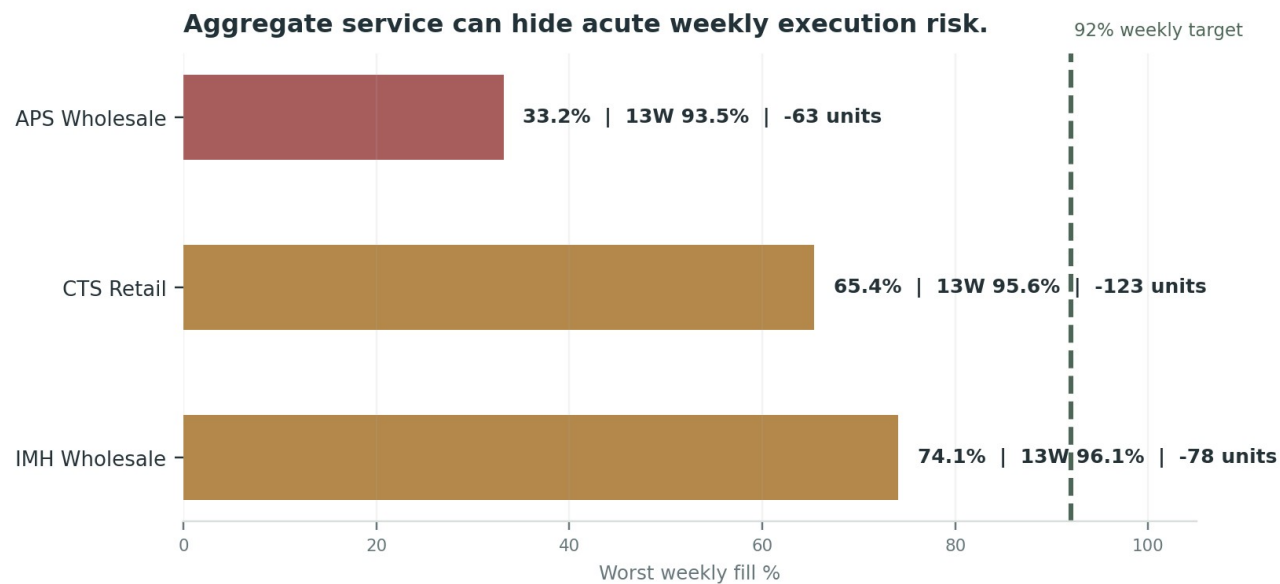
A useful planning model should be allowed to tell me that a factor is less important than I expected.

FORECAST EVIDENCE

CHAMPION MIX	WAPE RANGE	BIAS RANGE
8 ETS	8.4%-21.7%	-3.1% to +2.4%
1 baseline 0 SARIMA	ECOM/Retail 8-11% Wholesale 18-22%	all nine series

98.8% portfolio fill looked healthy. Weekly execution told a different story.

The aggregate KPI was not wrong - it was incomplete.



PORTFOLIO BASE FILL	P1 SERIES	WORST WEEKLY FILL	LARGEST WEEKLY GAP
98.77%	3 of 9	33.2%	-123 units
above 92% target	each misses weekly target twice	APS Wholesale	CTS Retail

What changed in my conclusion

If I had stopped at the 13-week portfolio fill rate, I would have concluded that service was healthy. Looking at weekly SKU x Channel execution exposed three concentrated failures that the aggregate ratio absorbed.

This is the planning lesson I would carry forward: manage the weeks that break the plan, not only the averages that summarize it.

The immediate service problem was concentrated. The coverage problem was portfolio-wide.

A shared receipt gap helped explain why thin-buffer series failed first.



ACUTE WEEKLY RISK	FORWARD COVERAGE RISK	P1 ENDING WOS	P2 ENDING WOS
3 of 9	9 of 9	0.00	0.28-1.25
repeated service misses	end below 2.5-week policy	weeks	weeks

How I would read this as a planner

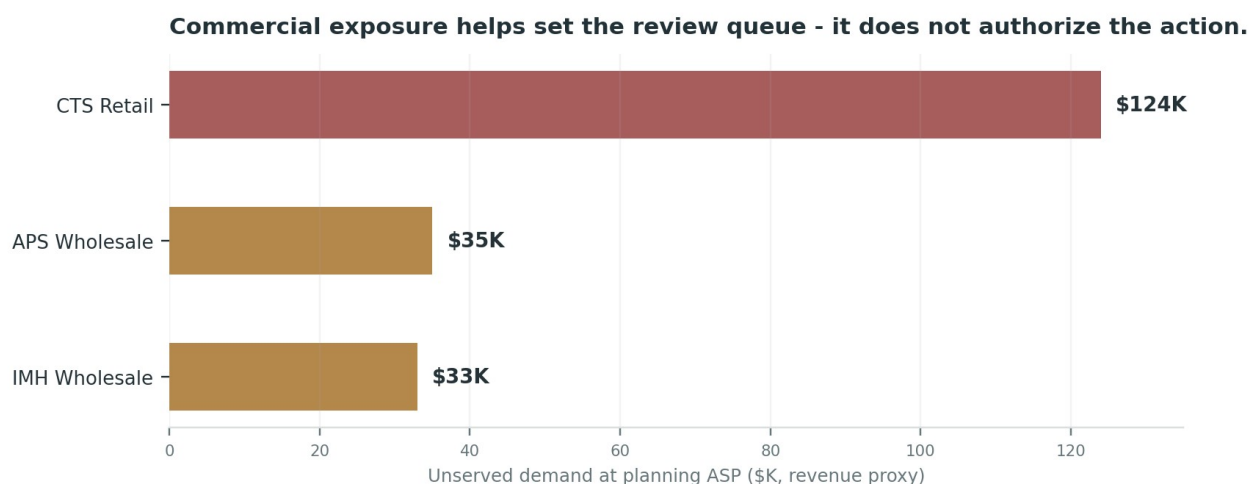
- Three series need immediate S&OE attention because weekly service already fails.
- All nine series deserve a forward-coverage conversation because every series ends the horizon thin.
- The ~7,000-unit gap to the 2.5-week policy is a coverage observation - not an automatic buy recommendation.

A policy gap tells me where to investigate. It does not tell me what I am operationally allowed to buy, expedite or transfer.

Prioritize the conversation; keep execution authorization with the planner.

DemandIQ detects and ranks the risk, but it does not pretend to know supplier, expedite or transfer feasibility.

P1 - ESCALATE 3 series APS-001 / Wholesale CTS-001 / Retail IMH-001 / Wholesale Why: repeated weekly service misses, thin buffers and measurable commercial exposure. Planner action: urgent S&OE review of receipts, expedite or transfer feasibility.	P2 - PROTECT 6 series Weekly service is currently healthy, but coverage finishes below the 2.5-week policy. Planner action: hold the current plan and re-check coverage weekly. Escalate only if service or coverage deteriorates. No automatic chase quantity is generated.
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BASE EXPOSURE ~\$192K revenue proxy - not margin	SEVERE EXPOSURE ~\$218K revenue proxy - not margin	CARRYING PROXY ~\$172K 13-week holding-cost proxy
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The S&OE questions I would bring to the room

- Can any committed receipt be pulled forward before the late-August gap?
- Which P1 series has a feasible expedite or transfer path?
- Should the 2.5-week coverage policy differ by channel?
- If capacity is constrained, do we prioritize service severity or commercial exposure?

My objective was to demonstrate planning judgment, not just forecasting skill.

DemandIQ is a portfolio simulation, but the reasoning pattern is the part I would carry into a real planning role.

Start with the business mechanism Before forecasting sales, I asked whether sales actually represented demand.	Test assumptions rather than defend them Weather was expected to matter. When scenario spread was modest, I shifted the focus instead of overselling it.
Look below aggregate KPIs 98.8% portfolio fill was not enough evidence that weekly execution was healthy.	Respect the limits of the model If feasibility is outside the model, I escalate the decision rather than manufacture an answer.

What I would build next

- Supplier / transfer execution feasibility so the system can move from escalation toward feasible action options.
- Point-in-time weather nowcasting for the first three forecast weeks.
- A separate new-product cold-start launch module for preseason planning.

The value of the case study is not that it reproduces a company's planning system. It shows how I would decompose the problem, govern the analysis, challenge the first answer, and move toward a planning decision.

Technical stack

Python | pandas | statsmodels | synthetic retail demand, inventory and weather simulation | expanding-window backtesting

What is modeled, what is observed, and what is intentionally outside scope.

The case study is designed to make the boundaries explicit rather than hide them.

Used in the planning workflow • reconstructed_demand_units as the forecasting target • week x SKU x Channel grain • opening inventory and committed receipts • weekly fill and weeks of supply • weather scenario layer • WAPE and bias for model governance • planning ASP / carrying-cost exposure proxies	Not treated as model inputs • true_demand_units (hidden synthetic evaluation only) • lost_demand_units • audit_hidden_* fields • weather_effect_pct / weather_factor • positive_spike_factor / negative_shock_factor • noise_factor These generator-only fields remain outside the forecasting feature set.
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DELIBERATE LIMITATIONS

- The economics are prioritization proxies, not accounting profit or margin forecasts.
- Supplier capacity, expedite feasibility, transfer feasibility, lead times and operational authorization are not modeled.
- Regions are aggregated before forecasting; the case study does not attempt store-level allocation.
- The mature-product 13-week horizon is an in-season scope choice. New-product preseason planning is a separate extension.

WHAT I WOULD WANT A HIRING MANAGER TO REMEMBER

1 Recover the signal stockouts can hide demand	2 Look below the aggregate weekly failures can hide inside healthy totals	3 Detect before you authorize planning judgment remains human
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Prepared as a portfolio case study. All figures shown are generated from the DemandIQ synthetic planning simulation.